

Electronic Engine Controls



The electronic engine controls consist of the following:

- Camshaft Position (CMP) Sensor
- Catalyst Monitor Sensor (CMS)
- Crankshaft Position (CKP) Sensor
- Cylinder Head Temperature (CHT) Sensor
- Fuel Rail Pressure (FRP) Sensor — 3.5L Gasoline Turbocharged Direct Injection (GTDI)
- Heated Oxygen Sensor (HO2S)
- Intake Air Temperature (IAT) Sensor — 3.5L GTDI
- Knock Sensor (KS)
- Manifold Absolute Pressure (MAP) /Intake Air Temperature 2 (IAT2) Sensor — 3.5L GTDI
- Mass Air Flow (MAF) Sensor — 3.7L, 5.0L (4V) and 6.2L (2V)
- PCM
- Turbocharger Boost Pressure (TCBP) /Charge Air Cooler (CAC) Sensor — 3.5L GTDI
- Variable Camshaft Timing (VCT) Oil Control Valve — 3.5L GTDI and 3.7L
- VCT Variable Force Solenoid — 5.0L (4V) and 6.2L (2V)

The CHT sensor:

- is mounted to the wall of the cylinder head and is not connected to any coolant passages.
- sends the PCM a signal indicating cylinder head temperature.

The CKP sensor:

- is mounted to the wall of the cylinder block.
- sends the PCM a signal indicating crankshaft position and engine speed.

The CMP sensor:

- is mounted to the cylinder head.
- sends the PCM a signal indicating camshaft position.

The CMS :

- is mounted to the catalytic converter.
- sends the PCM a signal indicating oxygen content in the exhaust after the exhaust leaves the catalytic converter.

The FRP sensor (3.5L GTDI):

- is mounted to the fuel rail.
- measures the pressure of the fuel rail and sends these signals to the PCM .

The HO2S :

- is mounted to the exhaust Y-pipe.
- sends the PCM a signal indicating oxygen content in the exhaust before the exhaust enters the catalytic converter.

The IAT sensor (3.5L GTDI):

- is mounted on the Air Cleaner (ACL) .

- sends the **PCM** a signal indicating the temperature of the air entering the engine.

The **KS** :

- is mounted to the wall of the cylinder block.
- sends a signal to the **PCM** indicating engine detonation.

The **MAF** sensor:

- is mounted on the **ACL** .
- sends the **PCM** a signal indicating mass airflow rate of air entering the engine.

The **MAP /IAT2** Sensor (3.5L **GTDI**):

- is mounted on the top of the intake manifold.
- is a combination sensor.
- measures manifold pressure and converts it to an electrical signal. This provides the **PCM** information on engine load.
- sends the **PCM** a signal indicating the temperature of the air entering the cylinders after passing through the **CAC** .

The **TCBP /CAC** Sensor (3.5L **GTDI** only):

- is mounted to the **CAC** outlet tube.
- is a combination sensor.
- measures turbocharger boost pressure and sends this information to the **PCM** .
- measures the temperature of the charge air exiting the **CAC** and sends this information to the **PCM** .

The Turbocharger Bypass Valve.(3.5L **GTDI** only):

NOTE: *The turbo bypass valve opens at key-on during cold start, this relieves pressure on the turbo system and prevents the turbos from spinning too fast during initial start-up. After 30 seconds the valve closes at which a popping noise can be heard. This is normal operation for the bypass valve.*

- is mounted to the **CAC** .
- prevents back flow through the turbocharger when the throttle is rapidly closed to avoid undesirable noise. The high pressure downstream of the turbocharger is vented back to the intake air upstream when the valve is open reducing pressure in the system.

The **VCT** Oil Control Valve (3.5L **GTDI** and 3.7L only)

- is mounted to the cylinder head.
- is an electronic solenoid that supplies force to actuate the **VCT** oil control valve. When the **PCM** transmits a signal based on the engine speed and load, the solenoid moves the **VCT** oil control valve activating either the advance circuit, retard circuit or hold position. The camshaft is then repositioned in relation to crankshaft timing to allow for optimum engine performance with lower emissions and reduced fuel consumption.

The **VCT** Variable Force Solenoid [5.0L (4V) and 6.2L (2V) only]:

- is mounted to the engine front cover.
- is an electronic solenoid that supplies force to actuate the **VCT** oil control valve. The **PCM** transmits a signal based on the engine speed and load, the solenoid will move the **VCT** oil control valve activating either the advance circuit, retard circuit or hold position. The camshaft is then repositioned in relation to crankshaft timing to allow for optimum engine performance with lower emissions and reduced fuel consumption.